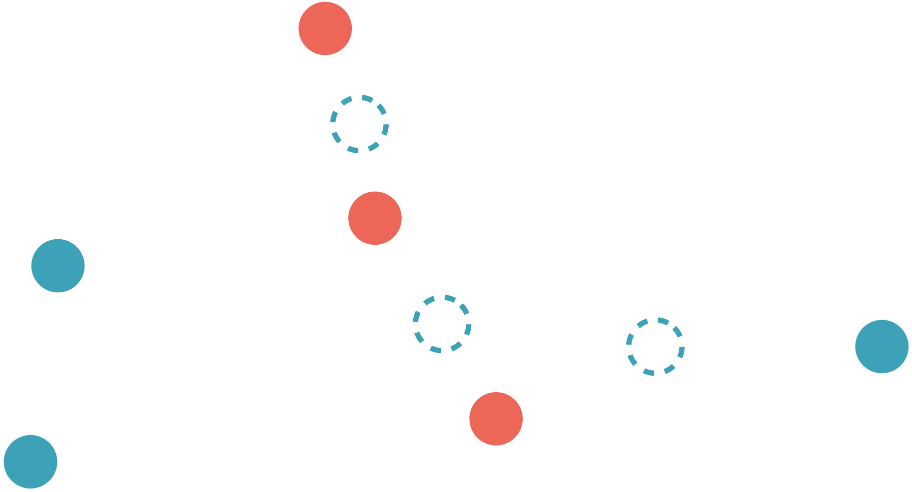
not a map

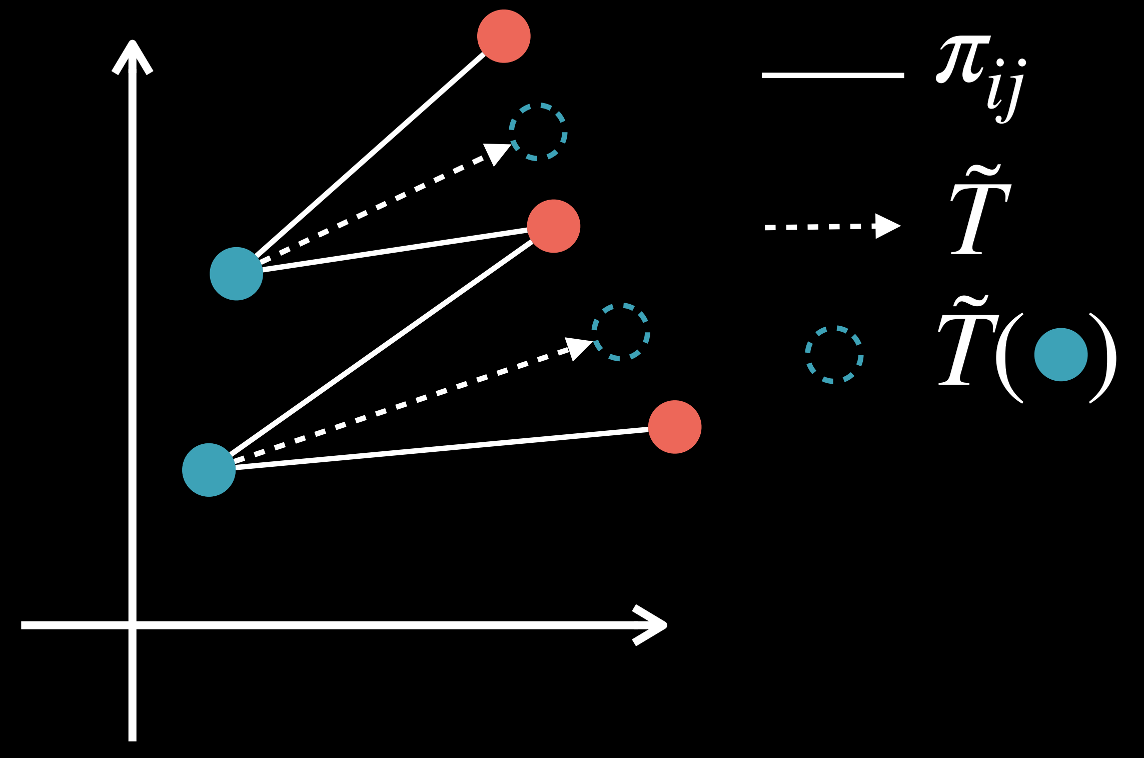
the barycentric projection

an approximate Monge map

no direct out-of-sample extension



Barycentric Projection



- Kantorovich formulation produces a coupling/plan, **not a map**.

- If π is a permutation matrix, it directly encodes a map $\pi_{ij} \neq 0 \Rightarrow T(x_i) = y_j$

- For dense couplings, we can still define a map via ***the barycentric projection***:
  $\tilde{T} : \mathbf{x}_i \mapsto \frac{1}{a_i} \sum_j \pi_{ij} \mathbf{y}_j$ (for general measures, $T : x \mapsto \int_{\mathcal{Y}} y \frac{d\pi(x,y)}{d\alpha(x)\, d\beta(y)} d\beta(y)$)

- In matrix form, for data $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $\mathbf{Y} \in \mathbb{R}^{m \times d}$, we can write $\tilde{T}(\mathbf{X}) = \mathrm{diag}(\mathbf{a})^{-1} \pi \mathbf{Y}$

- **Probabilistic view**: $\tilde{T}(\mathbf{x}) = \mathbb{E}_\pi[Y \mid X = \mathbf{x}]$, i.e., conditional expectation under π

- **Usefulness**: gives **an approximate Monge map** for tasks like domain adaptation, generative modeling, or style transfer.

- **Limitation**: only defined on the training/source samples $\mathbf{x}_i$; **no direct out-of-sample extension**.

Barycentric Projection